

B-LINE PUBLIC TRANSIT: A DATA DRIVEN ANALYSIS FOR SERVICE OPTIMIZATION

A Project

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DEDICATION

I would like to dedicate this project to the riders and operators of the B-Line public transit system, and to every individual who relies on it as a part of their daily life.

You wake up early to catch the first bus, travel across the city for work, school, and essential needs, and trust the system to connect you to opportunities and responsibilities.

You navigate changing schedules, long routes, and busy stops—not because it is easy, but because it is necessary. This project was built with you in mind, to better understand the patterns behind your journeys and the challenges you experience.

I hope this work contributes, even in a small way, to improving the reliability, accessibility, and effectiveness of the service you depend on every day.

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LIST OF SYMBOLS

Y_t = Total ridership on day t .

t = Time index representing day t ($t = 1, 2, \dots, 820$).

μ = Long-run (unconditional) mean of the time series.

c = Intercept term in the ARMA(1,1) model.

ϕ_1 = Autoregressive coefficient in the ARMA(1,1) model.

θ_1 = Moving-average coefficient in the ARMA(1,1) model.

ε_t = White-noise error term at time t , where $\varepsilon_t \sim WN(0, \sigma^2)$.

$WN(0, \sigma^2)$ = White-noise process with mean 0 and variance σ^2 .

σ^2 = Variance of the white-noise error term.

u_t = Autocorrelated error term in the GLSAR model.

ρ = First-order autoregressive coefficient in the GLSAR model.

β_0 = Intercept term in regression models.

β_1 = Slope coefficient representing the linear time trend in ridership.

β_p = Vector of regression coefficients for covariates in the GLSAR model.

X = Matrix of covariates including day-of-week, month, season, and route indicators.

B_i = Boardings per trip for trip i .

i = Index representing individual trips.

β_h = Regression coefficients for hour-of-day indicator variables.

γ_d = Regression coefficients for day-of-week indicator variables.

δ_r = Regression coefficients for route indicator variables.

ε_i = Error term in the trip-level regression model.

AIC = Akaike Information Criterion used for model selection.

BIC = Bayesian Information Criterion used for model selection.

ABSTRACT

This project presents a data-driven evaluation of the B-Line public transit system to identify patterns in rider demand and assess operational efficiency. Public transit agencies must balance service availability with resource constraints, yet demand often fluctuates across time and routes, leading to potential inefficiencies. This study examines how ridership varies over time, how service utilization differs across scheduled trips, and which routes or stops may be underutilized.

The analysis integrates validation-level boarding records spanning July 2022 through March 2025. Daily ridership totals were constructed to evaluate temporal demand patterns across days of the week, months, and seasons using time series modeling. Trip-level data were used to analyze boardings per trip through multivariable regression to assess differences in utilization by hour, day, and route. Route-level summaries were examined descriptively to identify persistent underutilization relative to service frequency.

Results indicate clear weekday and academic-season peaks in ridership, with reduced demand during weekends and summer months. Substantial variation in boardings per trip was observed across hours and routes, with several routes demonstrating consistently low utilization during off-peak periods. These findings suggest opportunities for schedule adjustments and targeted service optimization.

Overall, this study demonstrates how integrated transit data can support evidence-based planning decisions. The results provide actionable insights to improve operational efficiency while maintaining service accessibility for the community.

CHAPTER I

INTRODUCTION

Purpose of Project

Public transportation plays a critical role in ensuring mobility, reducing traffic congestion, and promoting sustainable urban living. For mid-sized communities such as Butte County, California, the B-Line transit system is a vital service connecting students, commuters, and residents to schools, workplaces, and essential services. However, like many transit systems, B-Line faces challenges in balancing efficiency, ridership demand, and resource allocation. The purpose of this project is to apply data science and analytics techniques to evaluate ridership patterns, identify inefficiencies, and provide actionable insights to support decision-making for transit optimization. By analyzing ridership and fare data, the project seeks to uncover trends that can guide improvements in scheduling, route planning, and fare policies.

Scope

This project focuses on ridership data from the B-Line transit system covering multiple years. Using statistical analysis, time series modeling, and visualization, the study examines temporal demand patterns, route utilization, and the impact of fare changes. The scope includes:

- Identifying fluctuations in ridership across times of day, days of the week, and seasons.
- Detecting underutilized routes and stops to highlight potential inefficiencies.

The intended audience includes transit administrators, city planners, policymakers, and researchers in transportation and data analytics. The project results are expected to contribute both to local decision-making for B-Line and to broader discussions about improving transit services in comparable urban areas.

Significance of the Project

The significance of this project lies in its ability to demonstrate how data-driven decision-making can improve public transportation systems. By applying analytical techniques such as time series forecasting, regression modeling, and anomaly detection, the study bridges academic research with real-world applications. The project is particularly relevant in the field of data science, where the integration of predictive modeling and visualization can directly inform policy and operational strategies. Ultimately, the project aims to enhance efficiency and sustainability within the B-Line system.

Limitations of the Project

Several limitations shape the scope of this study. The project relies on comprehensive historical ridership and transaction data provided by B-Line, which is largely complete and well-structured, with only minor inconsistencies such as mixed data types in a few fields that were resolved during preprocessing. Second, the analysis is limited to the variables provided in the datasets; while the dataset provides detailed transactional and ridership information, it lacks demographic attributes of individual riders, such as age, income, or occupation. This limits the ability to analyze how different population groups use

B-Line services or respond to fare and schedule changes. Although external factors such as weather conditions or community events were not integrated in the current analysis, they could be incorporated in future work since the dataset includes precise date and time information that would allow such enrichment. Finally, the recommendations do not account for practical constraints such as budget limitations or community-specific needs.

Definitions of Terms

- **B-Line Transit:**

The public bus system serving Butte County, California.

- **Ridership:**

The total number of passengers boardings recorded on the transit system.

- **Route Utilization:**

A measure of how effectively a bus route is utilized.

- **Time Series Modeling:**

Time series modeling is used in this project to analyze rider demand over time, specifically to identify long-term trends, seasonal patterns (e.g., recurring weekly or monthly fluctuations), and to generate forecasts for future ridership.

- **Trip:**

Defined a trip as a single scheduled bus run in the B-Line transit system, operating on a specific route, in a specific direction, and at a scheduled time, representing one execution of service. For example, a single trip in the dataset corresponds to Route 3 (Nord / East) operating at 9:00 AM on January 4, 2023, representing one complete scheduled bus run during which passenger boardings were recorded.

- **Peak Hours:**

Peak hours are defined as weekday morning commute periods from 7:00–9:00 AM and weekday afternoon commute periods from 3:00–6:00 PM. This time windows were selected based on exploratory analysis of the given data(validation dataset), which showed the highest average boardings per trip occurring during these periods. All other times, including weekends, are classified as off-peak.

CHAPTER II

LITERATURE REVIEW

Public transportation systems are central to sustainable urban mobility, yet they continue to face challenges related to efficiency, scheduling, and fluctuating ridership demand (Tuo, Ma, & Yang, 2013; Perumal, 2019). In response, researchers across the world have increasingly turned to optimization, modeling, and data-driven approaches to improve transit performance and planning.

A recurring theme in the literature is the importance of efficient network design in meeting diverse travel demands. Tuo et al. (2013) proposed a layout optimization algorithm that accounted for route length, passenger demand, and non-linear constraints, illustrating how well-designed transit networks can balance accessibility and operational efficiency. Although this work primarily targeted large urban settings, its principles of route rationalization and demand-based design provide a useful foundation for examining underutilized routes in smaller systems such as B-Line.

Beyond network structure, scholars have focused on service scheduling and frequency optimization to enhance both passenger experience and operator efficiency. Wagale et al. (2013) developed a demand- and travel-time responsive (DTR) model that adjusted headways in real time to minimize passenger waiting time while controlling operational costs. Similarly, Basak et al. (2019) advanced this concept by using data-driven and computational techniques—combining clustering analysis of seasonal delay patterns with genetic algorithms (GA) and particle swarm optimization (PSO) to improve on-time bus

performance. Their findings underscore the value of incorporating temporal and seasonal variations into service planning, a perspective that directly informs this project's focus on ridership fluctuations across different times and seasons.

However, while these optimization models demonstrate significant improvements in efficiency, they also rely heavily on real-time and high-resolution data, which may not be feasible for smaller or regional transit agencies. This presents a gap in the literature: there is limited research on how such methods can be adapted to mid-sized systems with access only to historical data.

Complementing these operational studies, broader analyses of ridership behavior have emphasized that declines in U.S. transit use since 2013 are less about service quality and more about external factors such as greater vehicle ownership, telecommuting, and the rise of ride-hailing services (FDOT, 2019). These studies reveal that improving public transit performance requires not only operational efficiency but also an understanding of social and behavioral shifts that influence ridership patterns.

Taking it together, the literature reveals several key insights and gaps. While optimization-based studies have advanced methods for scheduling and network design, their dependence on real-time data limits their applicability to smaller agencies. Meanwhile, behavioral research has deepened understanding of macro-level ridership trends but rarely links these to route-level operational decisions. In addition, prior research has shown that ridership and demand data are often modeled using count-based statistical approaches, such as Poisson regression, which are well-suited for non-negative and highly variable outcomes (McCullagh & Nelder, 1989; Cameron & Trivedi, 2013).

This project builds on prior research by applying data-driven analysis to historical ridership and transaction data from B-Line. By analyzing temporal demand fluctuations, route utilization, and fare elasticity, this study demonstrates how smaller transit systems can generate actionable insights to enhance efficiency and service planning without requiring real-time data infrastructure.

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CHAPTER III

METHODOLOGY

Introduction

This section presents the methodology used to address the study's research questions. The section begins with a detailed description of the datasets used, including their source, level of observation, time coverage, and key variables, followed by data preparation and preprocessing procedures. Analytical methods are then presented separately for each research question. For all research questions, exploratory data analysis (EDA), including summary statistics and visualizations, was conducted prior to inferential analysis to identify key patterns and guide the choice of appropriate statistical methods.

Data collection methods

Multiple datasets were obtained directly from Butte Regional Transit (B-Line), provided by the agency's operations and data management team as monthly CSV exports from their internal operational system. The analysis in this study is based on the validation (ridership) dataset.

The validation dataset, covering the period from July 2022 through March 2025, contains 134,548 individual boarding validation records across 26 variables. Each row represents a single passenger boarding event and includes timestamp information, route identifiers, vehicle number, and geographic attributes.

The route and stop dataset contain 20,738 records across 7 variables, providing geographic and structural information about bus stops in the B-Line system, including stop identifiers, stop names, and latitude/longitude coordinates.

The structure and variables of each dataset are described below:

- Validation data:
 - Level of observation: Each row represents a single passenger boarding event recorded when a rider validates a fare upon boarding a B-Line bus. It is important to note that the dataset includes only mobile fare validations and excludes boardings made using paper passes, student passes, or other non-digital payment methods, which may result in an underrepresentation of total ridership.
 - Used in:
 - *RQ1: How does rider demand fluctuate across different times of the day, days of the week, and seasons?*
 - ***RQ2:** How does service utilization, measured as ridership per trip, vary across different times of day in B-Line's current service?*
 - ***RQ3:** Which routes and stops are underutilized, and how can this be identified using ridership data?*
 - Key Variables Used:
 - timestamp — date and time of boarding validation

- route_name / route_number — route on which the boarding occurred
- trip_id — unique trip identifier
- vehicle_id — bus identifier
- stop_id — boarding stop identifier

Data preparation methods

During cleaning, duplicate records and incomplete rows (e.g., missing route IDs or timestamps) were removed, date and time fields were converted into consistent formats, and categorical variables such as route names and pass types were standardized. This process ensured uniform variable naming, consistent data types, and compatibility across all months and years, resulting in multiple continuous, standardized datasets that were analyzed separately, each suitable for time-based and comparative analyses.

All analyses were conducted using Python (version 3.11). Data preprocessing and manipulation were performed using the pandas library (McKinney, 2010) and NumPy (Harris et al., 2020). Time-series modeling, including ARIMA models and Augmented Dickey–Fuller stationarity tests, was implemented using the statsmodels package (Seabold & Perktold, 2010). Data visualizations were generated using Matplotlib (Hunter, 2007) and Seaborn (Waskom, 2021). All analyses were executed within a Jupyter Notebook computational environment to ensure reproducibility.

Research Question–Specific Data Preparation

The previous section described the overall structure and integration of the validation, transaction, and route/stop datasets. Each research question required additional data management and transformation tailored to its specific methodological approach. The following sections describe the research question–specific preprocessing procedures, including aggregation levels, variable construction, filtering criteria, and modeling datasets used for each analysis.

RQ1: How does rider demand fluctuate across different times of the day, days of the week, and seasons?

Converted timestamp variables into derived features: Hour of day was coded using a 24-hour clock ranging from 00 to 23, where 00 represents midnight. Day of week was coded from Sunday through Saturday based on the service date. Season was defined using astronomical boundaries: Winter (December 21–March 20), Spring (March 21–June 20), Summer (June 21–September 22), and Fall (September 23–December 20).

In the code appendix 1, `df` refers to the consolidated validation dataset, which serves as the primary dataset for addressing Research Question 1. The variable `df` is used as a standard abbreviation for “dataframe” within the Python environment.

Aggregated ridership counts at multiple levels, including hourly totals by route, daily ridership by day of week, monthly and seasonal summaries, and stop-level boardings. This multi-level aggregation enabled analysis of both short-term (hourly and daily) fluctuations and long-term (monthly and seasonal) ridership trends.

RQ2: How does service utilization, measured as ridership per trip, vary across different times of day in B-Line's current service?

In the B-Line validation dataset, the variable trip_id identifies a single bus run operating on a specific route, in a specific direction, and at a scheduled time. In this study, each unique trip_id is treated as one execution of service. (see Appendix 2)

Trip timestamps (see Appendix 3) were used to derive temporal variables, including hour of day and day of week. These variables are treated as categorical predictors in subsequent analyses to capture non-linear variation in utilization across the service schedule.

Service utilization is measured at the trip level as boardings per trip, defined as the total number of passenger boarding events observed on a single bus trip. Because each boarding event corresponds to one validation record, boardings per trip for trip i are computed as: $B_i = \sum_{j=1}^{n_i} 1$, Let B_i denote the number of boardings on trip i , where n_i represents the number of boarding records linked to trip i . where j indexes individual validation (boarding) events associated with trip i .

Route identifiers (route_id) were retained to capture spatial variation in ridership patterns across the B-Line network. The resulting trip-level dataset includes one observation per trip, with associated boardings per trip, hour of day, day of week, and route identifier, and serves as the analytical dataset for Research Question 2.

Prior to modeling, the trip-level dataset was prepared to ensure consistent and interpretable regression inputs. Hour of day, day of week, and route identifiers were treated as categorical variables to allow for non-linear variation in service utilization across the

schedule. Route identifiers were standardized to remove formatting inconsistencies arising from mixed data types. Descriptive checks were conducted to confirm the distribution of boardings per trip and to verify the set of categorical levels used in the analysis. These steps ensured a clean and stable dataset for multivariable regression estimation. (see Appendix 4).

Statistical analysis methods

Although the dataset consists of time-stamped observations, the choice of modeling approach is guided by the unit of analysis and the specific quantity being estimated, rather than the temporal ordering of the data alone (Wooldridge, 2016; Stock & Watson, 2019). Time-series methods are appropriate when the objective is to model temporal dependence, persistence, or dynamic evolution in an outcome over time (Box et al., 2015). In contrast, when the objective is to estimate systematic differences across observations conditional on explanatory variables, regression-based approaches are more appropriate (Wooldridge, 2016). Accordingly, multiple modeling frameworks are employed to align with the analytical goals of the study. For analyses involving aggregated ridership over time, where observations are sequential and potentially autocorrelated, time-series models are used to account for temporal dependence and identify patterns such as trend and seasonality (Box et al., 2015). For analyses conducted at the trip level, where the focus is on comparing average boardings across service conditions (e.g., hour, day, and route), multivariable regression models are used to estimate conditional mean differences without explicitly modeling dynamic time dependence (Wooldridge, 2016). While boardings per trip is a count variable, OLS regression is used in this context to estimate average differences in ridership

across service conditions. Given the large sample size and the focus on mean comparisons rather than modeling the full count distribution, OLS provides consistent estimates of conditional means. To address potential heteroskedasticity inherent in count data, heteroskedasticity-robust (HC3) standard errors are employed for inference. For analyses where modeling the distribution of count outcomes is central, such as estimating expected ridership for utilization comparisons, generalized linear models (e.g., Poisson regression) are employed to appropriately model non-negative count data (Cameron & Trivedi, 2013).

This approach ensures that each modeling framework is matched to the structure of the outcome variable and the research objective, providing interpretable and methodologically appropriate results across different levels of analysis.

RQ1: How does rider activity fluctuate across different times of the day, days of the week, and seasons?

To address Research Question 1, several analytical approaches were applied. First, descriptive statistics were computed, including the mean, variance, and distribution of ridership by hour, by day of the week, and by season. These summaries provided an initial understanding of how demand fluctuates over time. Visualization techniques were then used to further illustrate temporal patterns, including time series plots, calendar-based heatmaps, and seasonal decomposition.

Next, two time-series models were evaluated to analyze daily ridership patterns: an Autoregressive Moving Average model of order (1,1) (ARMA(1,1)) and a Generalized Least Squares model with AR(1) errors (GLSAR). Both model types are widely used in

transportation forecasting research (Box, Jenkins, & Reinsel, 2015; Hyndman & Athanasopoulos, 2018).

Autoregressive Moving Average ARMA(1,1)

For the ARMA(1,1) model, the Augmented Dickey–Fuller (ADF) test was first applied to assess whether a stationary model was appropriate. The test returned a p-value of 0.0015, indicating that the series was stationary (Said & Dickey, 1984). Given this result, the ARMA(1,1) model was subsequently evaluated as an appropriate time-series structure. ARMA models are suitable for stationary time series that display both autoregressive and moving-average behavior (Box et al., 2015), capturing short-term autocorrelation through dependence on both previous observations and prior forecast errors (Hyndman & Athanasopoulos, 2018).

For the ARMA(1,1) model, let Y_t denote the total ridership on day t , for $t = 1, 2, \dots, 820$, where $t = 1$ corresponds to January 2, 2023, and $t = 820$ corresponds to March 31, 2025. Because there is one aggregated ridership value per day, no additional index (such as i for individual riders or stops) is required. The ARMA(1,1) model fitted to this series is

$$Y_t = \mu + \phi_1 Y_{t-1} + \theta_1 \varepsilon_{t-1} + \varepsilon_t,$$

where $\varepsilon_t \sim WN(0, \sigma^2)$ is a white-noise error term, ϕ_1 is the autoregressive coefficient, and θ_1 is the moving-average coefficient. In the ARMA(1,1) model, the parameters include the intercept c , the autoregressive coefficient ϕ_1 , and the moving-average coefficient θ_1 . For an ARMA(1,1) process, the long-run (unconditional) mean of the series is given by $\mu = c/(1 - \phi_1)$. The terms ϕ_1 and θ_1 describe short-term temporal dependence in

the series, with ϕ_1 capturing the relationship between consecutive observations and θ_1 capturing the effect of past forecast errors. These parameters are estimated from the data during model fitting.

The ARMA model relies on several assumptions, and standard diagnostic procedures were used to evaluate whether these assumptions were met.

First, the assumption of stationarity was assessed using the Augmented Dickey–Fuller (ADF) test, which is commonly applied to determine whether the time series has a constant mean and variance over time.

Second, to evaluate whether the residuals were independently and identically distributed, residual diagnostics were examined using the Ljung–Box test for autocorrelation and visual inspection of residual plots, including histograms and Q–Q plots. These diagnostics help assess whether any structure remains unexplained by the model.

Finally, the assumption of linear dependence between observations was examined by inspecting the autocorrelation and partial autocorrelation functions (ACF and PACF) of the series prior to fitting the model, which informs the suitability of ARMA modeling.

Generalized Least Squares model with AR(1) errors (GLSAR)

As an alternative, a GLSAR model with AR(1) errors was also evaluated. This model was considered because preliminary plots suggested the presence of a linear trend in ridership over time (Diggle, Heagerty, Liang, & Zeger, 2002). GLSAR is appropriate when observations exhibit autocorrelation that violates ordinary least squares (OLS) assumptions (Greene, 2018), and the AR(1) error structure accounts for correlation between successive residuals, which is common in transportation time-series settings (Diggle et al., 2002). This

specification allows the model to capture a linear trend while explicitly modeling autocorrelated errors, and the AR(1) term ensures that serial correlation is not ignored (Greene, 2018; Diggle et al., 2002). Let Y_t denote total ridership on day t , for the same time period as the ARMA(1,1) model.

The GLSAR(AR1) model is where Y_t represents the same total daily ridership defined earlier (the aggregated number of boardings on day t), and X is a matrix of covariates including day-of-week variables (Tuesday–Sunday, with Monday as the reference), month variables (February–December, with January as the reference), seasonal indicators (Spring, Summer, Fall, with Winter as the reference), and an indicator for regional routes (with local routes as the reference category).

$$Y_t = \beta_0 + \beta_1 t + \beta_p X + u_t,$$

where the error term follows a first-order autoregressive process,

$$u_t = \rho u_{t-1} + \varepsilon_t, \varepsilon_t \sim WN(0, \sigma^2).$$

Here, β_0 is the intercept (baseline ridership level) and β_1 is the slope measuring any linear trend in ridership over time. A positive β_1 would indicate an increasing trend, while a negative β_1 would indicate a decreasing one. In this model, β_1 represents the linear trend component, allowing the model to capture potential increases or decreases in ridership over time. The statistical significance of β_1 is evaluated during model fitting to determine whether a meaningful trend is present. The term u_t represents the autocorrelated component of the error not captured by the linear trend, and ρ is the AR(1) coefficient

indicating how strongly today's error depends on yesterday's error. When $|\rho| < 1$, the error process is stationary. A positive ρ means that if ridership is higher than expected on one day, it tends to be higher than expected the next day as well (and similarly for lower-than-expected days). The residuals ε_t are assumed to be white noise, and GLSAR iteratively estimates ρ and the regression coefficients so that the residuals become approximately uncorrelated.

Model performance was evaluated using the Akaike Information Criterion (AIC), which balances model fit and complexity (Burnham & Anderson, 2004). The Bayesian Information Criterion (BIC) was also examined because it penalizes model complexity more strongly, especially in smaller datasets (Burnham & Anderson, 2004). Residual diagnostics—including the Ljung–Box test to detect autocorrelation—were used to assess model adequacy (Ljung & Box, 1978). For both the ARMA and GLSAR models, normality of residuals was examined using Q–Q plots following standard diagnostic guidelines (Hyndman & Athanasopoulos, 2018). The preferred model was selected based on minimizing AIC/BIC and producing residuals with no significant autocorrelation (Burnham & Anderson, 2004).

RQ2: How does service utilization, measured as ridership per trip, vary across different times of day in B-Line's current service?

To address Research Question 2, the statistical analysis focuses on explaining variation in service utilization at the trip level, measured as boardings per trip. Unlike Research Question 1, which examines aggregate ridership demand over time using time-series methods, Research Question 2 examines cross-sectional differences in boardings

per trip across service times and routes. Although trips occur sequentially over time, the estimand of interest is the conditional mean difference in utilization across hour, day of week, and route, rather than dynamic temporal dependence. Accordingly, a multivariable regression framework is used to model systematic differences in average boardings per trip. OLS estimation with heteroskedasticity-robust (HC3) (MacKinnon & White, 1985) standard errors is employed for inference.

To examine how boardings per trip vary across service conditions, a multivariable regression model is estimated with boardings per trip as the dependent variable. Hour of day and day of week are included as categorical variables to capture systematic differences in average trip-level boardings associated with scheduled service timing. Route indicators are included to account for persistent differences in baseline ridership across routes.

Formally, the model is specified as:

$$B_i = \beta_0 + \sum_h \beta_h C(\text{hour}_i) + \sum_d \gamma_d C(\text{day_of_week}_i) + \sum_r \delta_r C(\text{route}_i) + \varepsilon_i,$$

where B_i denotes boardings per trip for trip i ; $C(\text{hour}_i)$ and $C(\text{day_of_week}_i)$ represent categorical indicators for the time at which the trip operates; $C(\text{route}_i)$ represents route fixed effects; and ε_i is an error term. One category from each set of indicators is omitted as the reference group. The reference categories are 10:00 am for hour of day, this was selected as the reference category because it represents a mid-morning period that is neither among the highest nor lowest ridership hours. Using a central, non-extreme time point provides a stable baseline against which increases and decreases in boardings per trip at other hours can be interpreted. Monday for day of week, it is a typical weekday with full

service and commuter-oriented demand. Using it as a reference is common in transit and labor economics because it represents normal weekday travel. Route indicators are also included, with Route 14 serving as the reference route, it is described as a local route with moderate ridership, it is neither the highest- nor the lowest-ridership route, so it works well as a middle baseline. Accordingly, all estimated coefficients are interpreted relative to trips operating at 10:00 on Mondays on Route 14.

RQ3: Which routes and stops are underutilized, and how can this be identified using ridership data?

To address Research Question 3, the analysis focused on identifying routes that exhibit lower-than-expected ridership after accounting for temporal patterns in service demand. The objective was to determine whether certain routes operate below expected levels given when they run, rather than simply identifying routes with low raw ridership.

First, descriptive statistics were computed at the route level, including the mean, median, and standard deviation of boardings per trip, along with the total number of trips for each route. These summaries provided an initial understanding of variation in ridership across routes. Visualization techniques, including bar plots and ranked summaries of average boardings per trip, were used to compare route-level utilization patterns.

Next, a regression-based modeling approach was applied to estimate the expected number of riders per trip based on temporal factors. Because ridership per trip is a non-negative count variable, a Generalized Linear Model (GLM) with a Poisson distribution and log link function was used (McCullagh & Nelder, 1989). This approach is widely used for modeling count data in transportation settings.

Let Y_i denote the number of boardings for trip i , for $i = 1, 2, \dots, n$, where n represents the total number of trips observed. The Poisson regression model is specified as

$$Y_i \sim \text{Poisson}(\mu_i), \log(\mu_i) = \beta_0 + \beta_1 \text{DOW}_i + \beta_2 \text{Hour}_i,$$

where μ_i is the expected number of boardings for trip i , β_0 is the intercept, and the predictor variables include categorical indicators for day of the week (with Monday as the reference category) and hour of the day (with 10 AM as the reference category). The coefficients represent multiplicative effects on expected ridership.

This specification excludes route-level indicators so that expected ridership is determined solely by temporal factors. Although routes are known to exhibit different baseline ridership levels, route-specific effects would absorb these differences into the model and result in predictions that closely match each route's historical average. This would limit the ability to identify routes that are underperforming relative to general demand patterns.

By excluding route indicators, the model establishes a common temporal baseline for expected ridership across all trips. This allows for a meaningful comparison between observed and expected ridership at the route level, where deviations from this baseline can be interpreted as systematic under- or over-utilization. In this way, the approach distinguishes between routes that are inherently low-demand due to scheduling and those that perform below what would be expected given their operating conditions.

Model assumptions were evaluated using standard diagnostic procedures. The Poisson model assumes that the mean and variance of the response variable are equal; this

assumption was assessed by comparing the Pearson chi-square statistic to the residual degrees of freedom, a commonly used diagnostic for detecting overdispersion in count data models (McCullagh & Nelder, 1989; Cameron & Trivedi, 2013).

After fitting the model, predicted ridership values $\hat{\mu}_i$ were obtained for each trip. Residuals were computed as the difference between observed and predicted boardings, $e_i = Y_i - \hat{\mu}_i$. These trip-level predictions and residuals were then aggregated to the route level.

For each route r , the mean observed ridership $\bar{Y}_r$ and the mean predicted ridership $\bar{\mu}_r$ were calculated by averaging observed and predicted boardings across all trips assigned to that route.

An observed-to-expected ratio was defined as

$$\text{Utilization Ratio}_r = \frac{\bar{Y}_r}{\bar{\mu}_r}.$$

This ratio provides a standardized measure for comparing route-level performance. Routes were classified based on this ratio, and a threshold-based approach was used to identify routes with relatively low observed-to-expected values. Specifically, routes in the bottom 15th percentile of the utilization ratio distribution were classified as underutilized. This percentile-based threshold was chosen to provide a data-driven method for identifying the lowest-performing routes while avoiding arbitrary cutoffs, and to focus on a manageable subset of routes exhibiting the most substantial deviations from expected ridership.

Model performance was evaluated using log-likelihood and pseudo- R^2 measures. Residual-based diagnostics were also used to assess model adequacy.

CHAPTER IV

RESULTS

RQ1: How does rider activity fluctuate across different times of the day, days of the week, and seasons?

Descriptive Analysis:

The analytic sample for Research Question 1 draws on multiple data sources summarized in Table 1. The validation dataset spans July 2023 through March 2025 and contains 134,548 individual boarding records across 23 distinct B-Line routes. These boarding records were aggregated to the calendar-day level to construct daily ridership totals used for time-series analysis.

The transaction dataset includes 351,178 fare transaction records covering July 2022 through December 2022 and represents activity across 22 distinct routes. While not all transactions correspond to unique riders, this data provide complementary information on fare usage and service activity.

Finally, the map dataset contains 20,738 spatial records representing route and stop-level information across 103 distinct route-related entries. These data define the structural and geographic components of the transit network and support route-level analysis.

Table 1. Summary of Data Sources

<i>Dataset</i>	<i>Records(n)</i>	<i>Start Date</i>	<i>End Date</i>	<i>Distinct Routes</i>
<i>Validation Data</i>	134548	2023-01-02	2025-03-31	23
<i>Transaction Data</i>	351178	2022-07-01	2022-12-31	22
<i>Map Data</i>	20738	N/A	N/A	103

Hourly ridership (Figure 1) exhibits clear and systematic variation throughout the day. Average ridership increases sharply during early morning service hours, rising from 2.68 boardings at 5 AM (SD = 1.35; n = 123 days) to 12.47 boardings at 6 AM (SD = 3.76; n = 128 days), and peaking at 18.00 boardings at 7 AM (SD = 9.49; n = 151 days). Ridership declines thereafter, averaging 8.90 boardings at 8 AM (SD = 4.57; n = 165 days) and 7.05 boardings at 9 AM (SD = 3.39; n = 178 days). Evening ridership falls steadily after 5 PM, dropping below 5 boardings per hour after 6 PM. These patterns indicate strong commuter-driven demand concentrated in traditional workday travel windows.

Substantial variation is also observed across days of the week (Figure 2). Weekday ridership levels are relatively stable, with mean daily ridership ranging from 147.67 boardings on Fridays (SD = 31.99; n = 27 days) to 163.58 boardings on Wednesdays (SD = 26.10; n = 26 days). In contrast, weekend ridership is considerably lower. Saturdays average 51.00 boardings (SD = 12.18; n = 27 days), while Sundays average only 10.40 boardings (SD = 3.81; n = 25 days). The magnitude of the weekday–weekend difference suggests that B-Line ridership is primarily driven by weekday commuting and institutional travel patterns.

Figure 1. Hourly Rider Demand

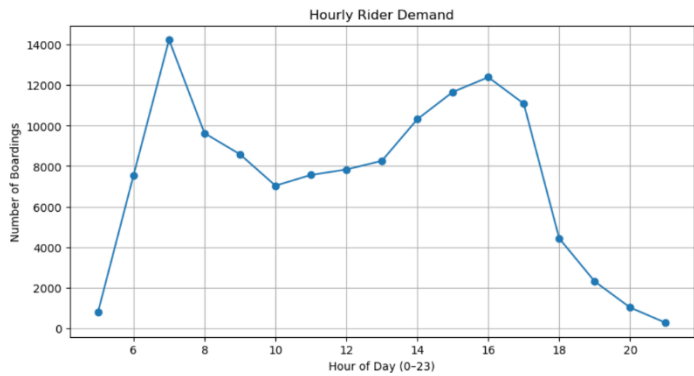
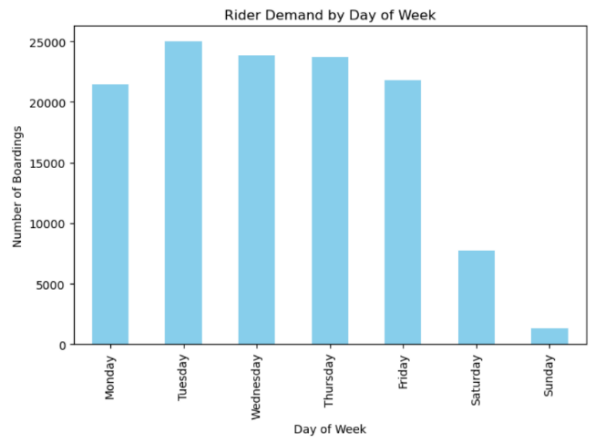


Figure 2. Rider Demand by Days



Monthly ridership patterns (Figure 3) show notable variation across the year. Ridership is highest during the early months, with March (mean = 20.64; SD = 63.07), February (mean = 19.89; SD = 63.44), and January (mean = 19.12; SD = 59.37) exhibiting the greatest average daily boardings. Ridership begins to decline in April (mean = 14.30; SD = 55.58) and May (mean = 14.94; SD = 56.88), followed by a more pronounced decrease during the summer months of June (mean = 11.45; SD = 43.87) and July (mean = 9.73; SD = 38.68), with July representing the lowest mid-year demand. Ridership recovers in August (mean = 13.75; SD = 52.73) and continues to increase through September (mean = 14.80; SD = 58.22) and October (mean = 15.82; SD = 60.28), before declining sharply again in November (mean = 6.41; SD = 37.06) and December (mean = 6.88; SD = 39.67).

The large standard deviations across months indicate substantial day-to-day variability in ridership within each period. Overall, the pattern suggests higher ridership during active academic months and reduced demand during summer and late-year periods,

likely reflecting seasonal academic schedules, employment patterns, and holiday-related travel behavior.

Figure 3. Rider Demand by Months

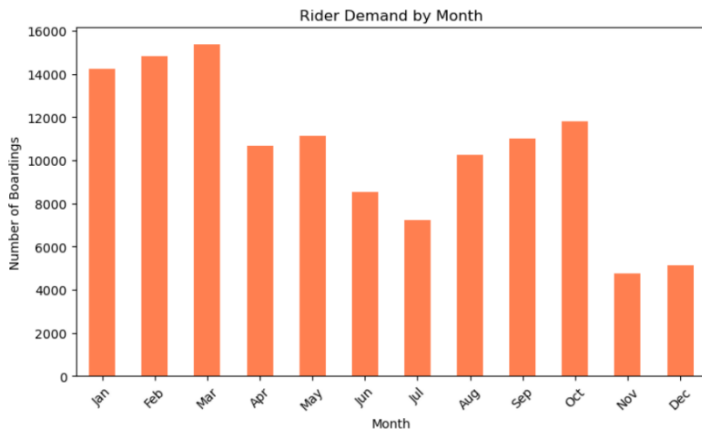
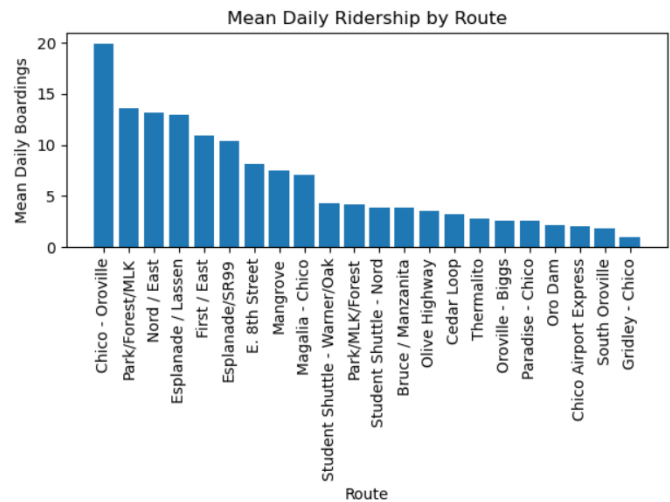


Figure 4. Mean Daily Ridership by Route

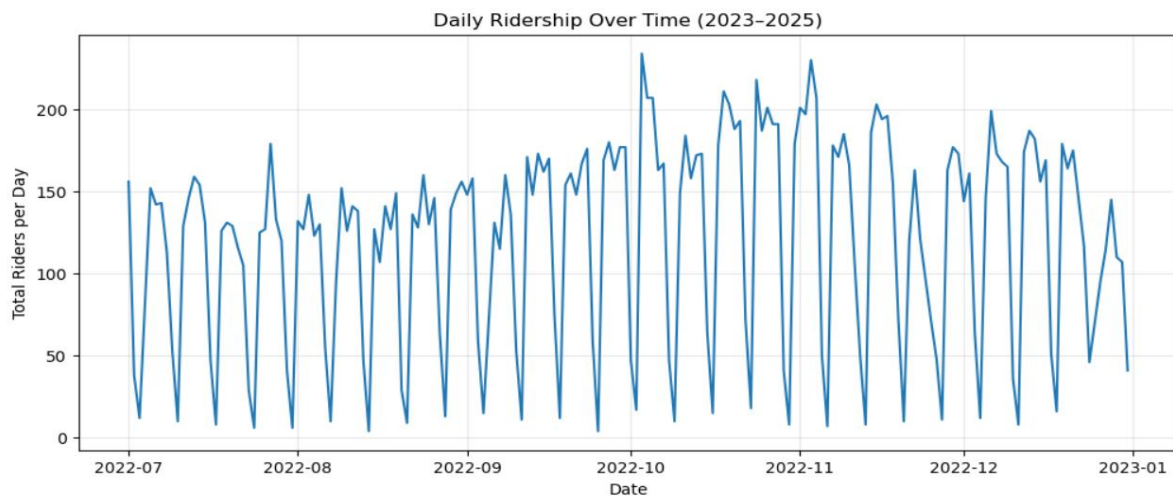


Substantial variation is observed across routes (Figure 4). Average daily ridership is highest on the Chico – Oroville route (mean = 19.93 boardings; SD = 11.24; n = 180 days), followed by Park/Forest/MLK (mean = 13.59; SD = 7.76; n = 163 days) and Nord / East (mean = 13.17; SD = 8.41; n = 157 days). In contrast, lower-utilization routes include Cedar Loop (mean = 3.20; SD = 3.02; n = 104 days), Olive Highway (mean = 3.51; SD = 1.78; n = 130 days), and Bruce / Manzanita (mean = 3.89; SD = 2.16; n = 145 days). The relatively large standard deviations for higher-ridership routes indicate substantial day-to-day variability in demand. These differences suggest meaningful heterogeneity in route-level utilization across the B-Line network.

Modelling Results:

Figure 5 displays total daily ridership from July 2022 through early 2023. Each point represents the total number of boarding validations recorded on a given calendar day. The figure exhibits a pronounced weekly cyclical pattern, with recurring sharp declines corresponding primarily to weekend days—especially Sundays—when ridership is substantially lower than on weekdays.

Figure 5. Daily Ridership over time



Weekday ridership levels remain consistently elevated relative to weekends, producing the repeated peaks and troughs observed throughout the series. This visible periodic structure suggests short-term dependence in daily ridership totals.

Time-series modeling aligns with these descriptive observations. The ARMA (1,1) model output (Figure 6) and diagnostic plots (Figure 7) show significant short-term dependence but no evidence of a sustained long-term upward or downward trend. In the

fitted ARMA (1,1) model, the estimated parameters were $c = 169.10$, $\phi_1 = -0.0091$, and $\theta_1 = 0.6453$.

The constant term c represents the baseline level of daily ridership around which short-term fluctuations occur. Using the estimated parameters, the implied long-run mean ridership is approximately 168 riders per day, indicating that B-Line demand fluctuates around a relatively stable average during the study period.

The autoregressive parameter ϕ_1 is close to zero, indicating that daily ridership is not strongly influenced by the previous day's ridership level once short-term fluctuations are considered. In practical terms, this suggests that high or low ridership on one day does not directly carry over in a sustained way to the next day.

In contrast, the moving-average parameter θ_1 is positive and statistically significant, indicating that unexpected changes (shocks) in ridership tend to carry over into the following day. This suggests that short-term disruptions or surges in demand influence ridership patterns in the immediate subsequent period.

Figure 6. ARMA (1,1) Model Output for Daily Ridership (2023–2025)

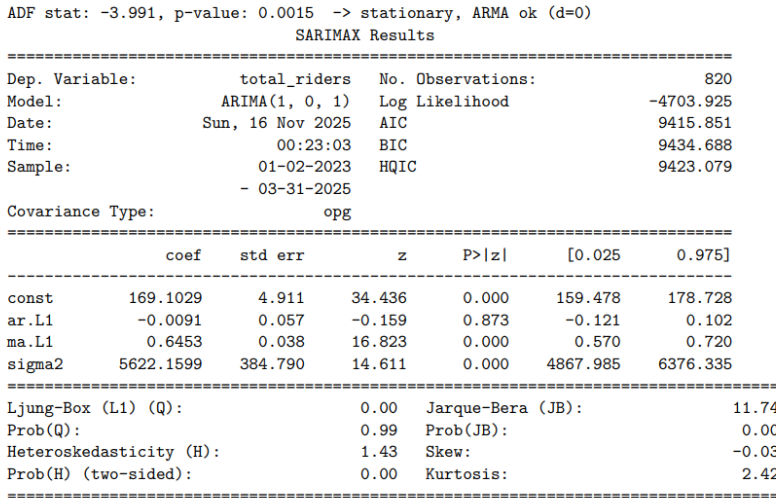
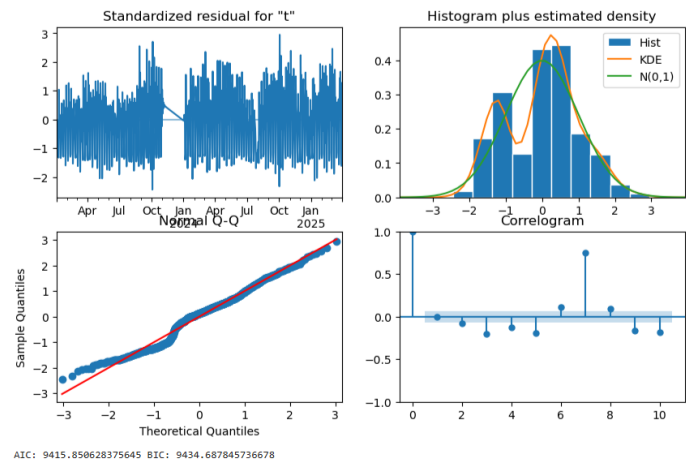


Figure 7. ARMA (1,1) Diagnostic Plots



To assess model adequacy, several diagnostic checks were performed on the residuals of the ARMA (1,1) model (Figure 7). The standardized residual plot indicates that residuals fluctuate randomly around zero with no sustained systematic pattern, suggesting that the model captures the primary temporal dependence in daily ridership. Although a brief deviation appears in early 2024, this fluctuation is temporary and does not persist across the series.

The histogram and kernel density estimate of residuals (Figure 7, top right) show an approximately symmetric distribution centered at zero. While minor deviations from perfect normality are observed in the tails, the residuals broadly conform to the normality assumption required for inference in ARMA models. This conclusion is further supported by

the Q–Q plot (Figure 7, bottom left), where most residual quantiles align closely with the theoretical normal reference line, with only slight departures in the lower extreme.

Finally, the autocorrelation function (ACF) of the residuals (Figure 7, lower right) plots residual autocorrelation coefficients against lag length, where the horizontal axis represents lag in days. Most autocorrelation coefficients fall within the 95% confidence bounds, indicating that the ARMA (1,1) model has largely removed serial dependence from the series. However, a noticeable spike appears at lag 7, corresponding to a one-week interval. This suggests the presence of residual weekly seasonality, likely reflecting systematic differences between weekday and weekend ridership patterns.

Despite this localized weekly dependence, the absence of significant autocorrelation at other lags indicates that the model captures the primary short-term dynamics of daily ridership. Collectively, these diagnostics suggest that the ARMA (1,1) specification provides a reasonable representation of the overall temporal structure.

Regression results (Table 2) show that day-of-week and month significantly influence total daily ridership, with Monday and January serving as reference categories. Relative to Mondays, ridership is significantly higher on Tuesdays ($\beta = 27.55$, $p < .0001$), Wednesdays ($\beta = 19.24$, $p < .0001$), and Thursdays ($\beta = 19.92$, $p < .0001$). These results indicate that midweek ridership exceeds Monday levels by approximately 19–28 riders per day, holding month constant. In contrast, weekend effects are substantially negative. Saturdays average approximately 133.59 fewer riders than Mondays ($p < .0001$), while Sundays average 193.38 fewer riders ($p < .0001$). These large negative coefficients confirm the strong weekend decline observed in the descriptive analysis. The Friday coefficient is small and not

statistically significant ($\beta = -1.96$, $p = .70$), suggesting that Friday ridership does not differ meaningfully from Monday after controlling for monthly variation.

Monthly indicators reveal meaningful within-year variation relative to January. Ridership is significantly higher in February ($\beta = 18.96$, $p < .0001$), March ($\beta = 13.69$, $p = .01$), April ($\beta = 25.30$, $p < .0001$), May ($\beta = 26.94$, $p < .0001$), August ($\beta = 14.67$, $p = .02$), September ($\beta = 39.74$, $p < .0001$), October ($\beta = 33.41$, $p < .0001$), and December ($\beta = 19.93$, $p = .01$). Among these, September and October exhibit the largest positive effects, with daily ridership exceeding January levels by approximately 33–40 riders per day. In contrast, July shows a significant decline relative to January ($\beta = -28.26$, $p < .0001$), consistent with reduced summer travel demand. June ($\beta = -9.89$, $p = .11$) and November ($\beta = 14.05$, $p = .08$) are not statistically significant at conventional levels, indicating no meaningful difference from January during those months.

Table 2. Regression results for daily ridership

Variable	Coefficient	Std. Error	t-value	p-value	CI Lower (95%)	CI Upper (95%)
<i>Intercept</i>	192.17	5.20	36.94	< 0.0001	181.96	202.38
<i>Friday</i>	-1.96	5.16	-0.38	0.70	-12.08	8.16
<i>Saturday</i>	-133.59	5.14	-25.98	<0 .0001	-143.68	-123.49
<i>Sunday</i>	-193.38	5.15	-37.52	< 0.0001	-203.50	-183.27
<i>Thursday</i>	19.92	5.18	3.84	< 0.0001	9.75	30.09
<i>Tuesday</i>	27.55	5.15	5.34	< 0.0001	17.43	37.67
<i>Wednesday</i>	19.24	5.17	3.72	< 0.0001	9.09	29.39
<i>February</i>	18.96	5.67	3.34	< 0.0001	7.82	30.09
<i>March</i>	13.69	5.55	2.47	0.01	2.80	24.58
<i>April</i>	25.30	6.25	4.05	< 0.0001	13.03	37.57
<i>May</i>	26.94	6.25	4.31	< 0.0001	14.67	39.22
<i>June</i>	-9.89	6.25	-1.58	0.11	-22.16	2.38
<i>July</i>	-28.26	6.32	-4.47	< 0.0001	-40.67	-15.86
<i>August</i>	14.67	6.25	2.35	0.02	2.40	26.95

September	39.74	6.32	6.29	< 0.0001	27.34	52.14
October	33.41	6.19	5.40	< 0.0001	21.26	45.56
November	14.05	8.01	1.75	0.08	-1.68	29.78
December	19.93	7.91	2.52	0.01	4.40	35.46

The GLSAR model summary (Figure 8) and residual analyses (Figures 9&10) similarly indicate that a linear time trend is not statistically significant, reinforcing that ridership does not follow a long-run increasing or decreasing trajectory over the study period. Consistent with this, the estimated linear trend coefficient in the GLSAR model was $\beta_1 = 0.0295$ (SE = 0.020, 95% CI: -0.010 to 0.069, $p = 0.147$), indicating that the daily change in ridership is small and not statistically different from zero, and therefore does not support the presence of a meaningful long-term linear trend.

Figure 8. GLSAR model summary

GLSAR Regression Results						
=====						
Dep. Variable:	total_riders	R-squared:	0.003			
Model:	GLSAR	Adj. R-squared:	0.001			
Method:	Least Squares	F-statistic:	2.107			
Date:	Sun, 16 Nov 2025	Prob (F-statistic):	0.147			
Time:	00:23:32	Log-Likelihood:	-4758.5			
No. Observations:	819	AIC:	9521.			
Df Residuals:	817	BIC:	9530.			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	157.1858	9.631	16.321	0.000	138.282	176.090
time	0.0295	0.020	1.452	0.147	-0.010	0.069
=====						
Omnibus:	64.639	Durbin-Watson:	1.652			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	27.911			
Skew:	-0.238	Prob(JB):	8.69e-07			
Kurtosis:	2.231	Cond. No.	950.			
=====						
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correct						
AIC: 9520.904067343015 BIC: 9530.320235510722						

Figure 9. GLSAR (AR1) Residuals

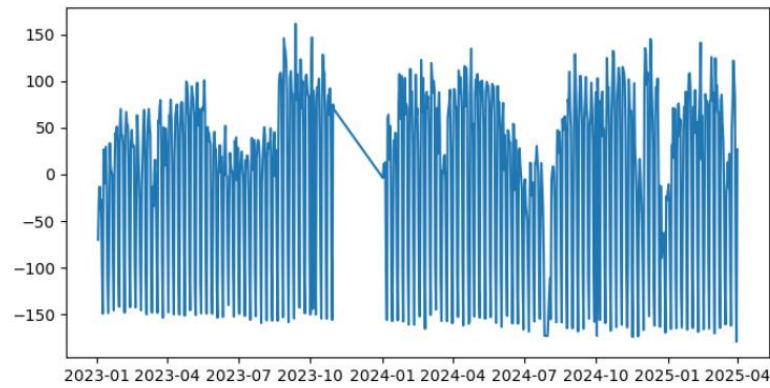
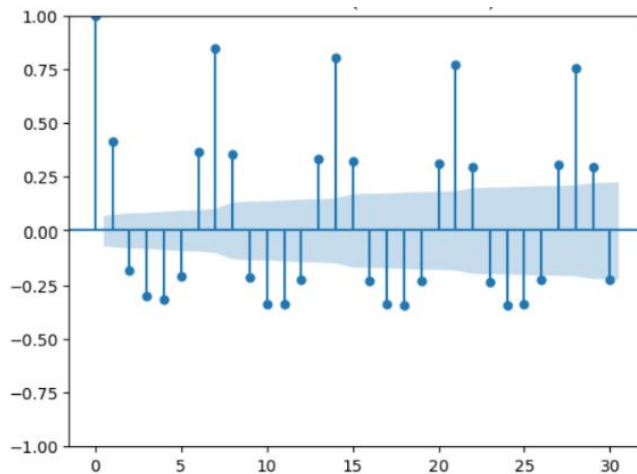


Figure 10. Residual ACF (GLSAR Model)



A multivariable ordinary least squares (OLS) regression (Figure 11) was estimated to examine variation in daily ridership as a function of day-of-week and month indicators (Table 1). The model is statistically significant overall (F-test $p < .0001$) and explains a substantial proportion of the variation in daily ridership (adjusted $R^2 = 0.833$). Using Monday as the reference category, Saturday ridership is estimated to be 133.59 riders lower per day (95% CI: -143.68, -123.49; $p < .0001$), and Sunday ridership is 193.38 riders lower per day (95%

CI: -203.50, -183.27; $p < .0001$), indicating pronounced weekend declines relative to weekdays. In contrast, Tuesday ($\beta = 27.55$; 95% CI: 17.43, 37.67; $p < .0001$), Wednesday ($\beta = 19.24$; 95% CI: 9.10, 29.39; $p < .0001$), and Thursday ($\beta = 19.92$; 95% CI: 9.75, 30.09; $p < .0001$) exhibit significantly higher ridership compared to Monday, while Friday does not differ significantly ($\beta = -1.96$; 95% CI: -12.08, 8.16; $p = .70$).

Monthly indicators reveal substantial variation across the calendar year. Relative to the reference month (January), ridership is significantly higher in February ($\beta = 18.96$; 95% CI: 7.82, 30.09; $p < .0001$), March ($\beta = 13.69$; 95% CI: 2.80, 24.58; $p = .01$), April ($\beta = 25.30$; 95% CI: 13.03, 37.57; $p < .0001$), May ($\beta = 26.94$; 95% CI: 14.67, 39.22; $p < .0001$), August ($\beta = 14.67$; 95% CI: 2.40, 26.95; $p = .02$), September ($\beta = 39.74$; 95% CI: 27.34, 52.14; $p < .0001$), October ($\beta = 33.41$; 95% CI: 21.26, 45.56; $p < .0001$), and December ($\beta = 19.93$; 95% CI: 4.40, 35.46; $p = .01$). In contrast, June ($\beta = -9.89$; 95% CI: -22.16, 2.38; $p = .11$) and November ($\beta = 14.05$; 95% CI: -1.68, 29.78; $p = .08$) do not differ significantly from January, while July shows significantly lower ridership ($\beta = 28.26$; 95% CI: 40.67, 15.86; $p < .0001$).

Figure 11. Multivariable ordinary least squares (OLS) regression

OLS Regression Results						
=====						
Dep. Variable:	total_riders	R-squared:	0.837			
Model:	OLS	Adj. R-squared:	0.833			
Method:	Least Squares	F-statistic:	219.9			
Date:	Wed, 08 Apr 2026	Prob (F-statistic):	8.97e-273			
Time:	14:41:35	Log-Likelihood:	-3748.2			
No. Observations:	745	AIC:	7532.			
Df Residuals:	727	BIC:	7615.			
Df Model:	17					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	192.1708	5.202	36.940	0.000	181.958	202.384
C(day_of_week, Treatment(reference="Monday"))[T.Friday]	-1.9574	5.155	-0.380	0.704	-12.079	8.164
C(day_of_week, Treatment(reference="Monday"))[T.Saturday]	-133.5869	5.142	-25.978	0.000	-143.683	-123.491
C(day_of_week, Treatment(reference="Monday"))[T.Sunday]	-193.3840	5.154	-37.523	0.000	-203.502	-183.266
C(day_of_week, Treatment(reference="Monday"))[T.Thursday]	19.9154	5.180	3.845	0.000	9.746	30.085
C(day_of_week, Treatment(reference="Monday"))[T.Tuesday]	27.5500	5.155	5.344	0.000	17.430	37.670
C(day_of_week, Treatment(reference="Monday"))[T.Wednesday]	19.2404	5.168	3.723	0.000	9.095	29.386
C(month)[T.2]	18.9557	5.672	3.342	0.001	7.819	30.092
C(month)[T.3]	13.6934	5.547	2.469	0.014	2.804	24.583
C(month)[T.4]	25.2992	6.251	4.047	0.000	13.027	37.572
C(month)[T.5]	26.9424	6.252	4.310	0.000	14.669	39.216
C(month)[T.6]	-9.8909	6.252	-1.582	0.114	-22.164	2.382
C(month)[T.7]	-28.2640	6.318	-4.474	0.000	-40.668	-15.861
C(month)[T.8]	14.6736	6.251	2.347	0.019	2.402	26.945
C(month)[T.9]	39.7380	6.317	6.291	0.000	27.337	52.139
C(month)[T.10]	33.4084	6.190	5.397	0.000	21.256	45.561
C(month)[T.11]	14.0482	8.011	1.754	0.080	-1.680	29.776
C(month)[T.12]	19.9280	7.909	2.520	0.012	4.400	35.456
=====						
Omnibus:	242.772	Durbin-Watson:	1.192			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1507.075			
Skew:	-1.321	Prob(JB):	0.00			
Kurtosis:	9.448	Cond. No.	11.7			
=====						
Notes:						
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.						

RQ 2:

Modelling Results:

Service utilization was analyzed at the trip level using boardings per trip as the primary outcome. The final analytical dataset included 9,469 trips, each representing a single scheduled bus run. Boardings per trip exhibited substantial variability across trips, with a mean of 13.83 riders (SD = 22.92) and a median of 6 riders. The IQR extended from 2 riders at the 25th percentile to 15 riders at the 75th percentile, while the maximum observed value reached 398 riders on a single trip. The large standard deviation relative to the mean, along with the wide range of observed boardings per trip, indicates substantial heterogeneity

in utilization across both the service schedule and routes. This suggests that ridership is not evenly distributed but instead varies considerably across individual trips. Many trips operate with relatively low boardings, while a smaller number of trips carry substantially higher passenger volumes, contributing to the observed variability. This pattern is consistent with the positively skewed distribution of boardings per trip, where a limited number of high-ridership trips increase the mean while the median remains comparatively lower. Such variability reflects differences in service timing, route characteristics, and demand patterns, indicating that utilization is concentrated among specific segments of the network rather than being uniform across all trips.

Table 3. OLS Regression Results: Effects of Hour, Day of Week, and Route

Variable	β	SE	z	p-value	[0.025	0.975]
<i>Intercept</i>	23.84	1.75	13.60	<0.0001	20.40	27.27
<i>C(hour)[T.5]</i>	59.87	21.01	2.85	<0.0001	18.68	101.03
<i>C(hour)[T.6]</i>	12.41	2.16	5.74	<0.0001	8.17	16.64
<i>C(hour)[T.7]</i>	10.31	1.49	6.91	<0.0001	7.39	13.24
<i>C(hour)[T.8]</i>	0.75	1.01	0.74	0.46	-1.23	2.72
<i>C(hour)[T.9]</i>	-0.57	0.89	-0.64	0.52	-2.31	1.17
<i>C(hour)[T.11]</i>	-1.16	0.85	-1.37	0.17	-2.81	0.50
<i>C(hour)[T.12]</i>	1.29	0.98	1.33	0.18	-0.61	3.21
<i>C(hour)[T.13]</i>	1.47	1.02	1.45	0.15	-0.52	3.47
<i>C(hour)[T.14]</i>	4.48	1.24	3.61	<0.0001	2.05	6.91
<i>C(hour)[T.15]</i>	3.10	1.07	2.90	<0.0001	1.01	5.20
<i>C(hour)[T.16]</i>	2.85	1.03	2.76	<0.0001	0.83	4.87
<i>C(hour)[T.17]</i>	0.37	1.02	0.37	0.71	-1.63	2.38
<i>C(hour)[T.18]</i>	-4.71	0.90	-5.26	<0.0001	-6.46	-2.96
<i>C(hour)[T.19]</i>	-5.69	0.94	-6.08	<0.0001	-7.52	-3.85
<i>C(hour)[T.20]</i>	-7.61	0.91	-8.39	<0.0001	-9.39	-5.83
<i>C(hour)[T.21]</i>	-8.02	1.56	-5.14	<0.0001	-11.08	-4.97

<i>C(day_of_week)[T.Tuesday]</i>	-3.89	0.89	-4.36	<0.0001	-5.65	-2.14
<i>C(day_of_week)[T.Wednesday]</i>	-6.70	0.87	-7.68	<0.0001	-8.42	-4.99
<i>C(day_of_week)[T.Thursday]</i>	-7.01	0.91	-7.66	<0.0001	-8.80	-5.22
<i>C(day_of_week)[T.Friday]</i>	-12.45	0.76	-16.46	<0.0001	-13.93	-10.97
<i>C(day_of_week)[T.Saturday]</i>	-12.79	0.82	-15.51	<0.0001	-14.40	-11.17
<i>C(day_of_week)[T.Sunday]</i>	-9.73	2.25	-4.34	<0.0001	-14.13	-5.33
<i>C(route_id)[T.15]</i>	-6.73	1.56	-4.32	<0.0001	-9.78	-3.68
<i>C(route_id)[T.16]</i>	-3.63	1.64	-2.22	0.02	-6.84	-0.42
<i>C(route_id)[T.17]</i>	-0.62	2.11	-0.29	0.77	-4.76	3.52
<i>C(route_id)[T.2]</i>	-6.40	1.54	-4.16	<0.0001	-9.42	-3.39
<i>C(route_id)[T.20]</i>	4.63	1.84	2.52	0.01	1.02	8.23
<i>C(route_id)[T.24]</i>	-5.96	1.98	-3.01	<0.0001	-9.85	-2.07
<i>C(route_id)[T.25]</i>	-12.53	1.61	-7.77	<0.0001	-15.69	-9.37
<i>C(route_id)[T.26]</i>	-10.28	1.76	-5.86	<0.0001	-13.71	-6.84
<i>C(route_id)[T.27]</i>	-13.13	1.60	-8.20	<0.0001	-16.27	-9.99
<i>C(route_id)[T.3]</i>	-4.50	1.58	-2.85	<0.0001	-7.60	-1.41
<i>C(route_id)[T.30]</i>	-6.36	1.81	-3.52	<0.0001	-9.90	-2.82
<i>C(route_id)[T.32]</i>	-14.47	3.50	-4.14	<0.0001	-21.33	-7.61
<i>C(route_id)[T.4]</i>	-4.99	1.53	-3.27	<0.0001	-7.98	-1.99
<i>C(route_id)[T.40]</i>	-2.87	2.14	-1.34	0.18	-7.06	1.32
<i>C(route_id)[T.41]</i>	7.63	2.66	2.87	<0.0001	2.42	12.83
<i>C(route_id)[T.5]</i>	-5.04	1.56	-3.24	<0.0001	-8.09	-1.99
<i>C(route_id)[T.52]</i>	-10.95	2.19	-5.00	<0.0001	-15.24	-6.66
<i>C(route_id)[T.7]</i>	-6.84	1.78	-3.84	<0.0001	-10.32	-3.35
<i>C(route_id)[T.8]</i>	-8.87	1.97	-4.50	<0.0001	-12.73	-5.01
<i>C(route_id)[T.9]</i>	-8.76	1.73	-5.08	<0.0001	-12.14	-5.38
<i>C(route_id)[T.9C]</i>	-9.36	2.02	-4.64	<0.0001	-13.32	-5.40

Table 3 presents OLS estimates of boardings per trip as a function of hour of day, day of week, and route indicators, with HC3 robust standard errors. The reference categories are 10:00 for hour of day, Monday for day of week, and Route 14. The model is statistically significant overall (F-test $p < 0.001$), indicating that the included predictors jointly improve model fit relative to an intercept-only model. The model explains approximately 13.6% of the variation in boardings per trip (adjusted $R^2 = 0.132$), reflecting meaningful but partial explanatory power given the heterogeneity in trip-level ridership outcomes.

Substantial variation in boardings per trip is observed across hours of operation (Table 3). Relative to the 10:00 reference hour, trips operating at 6:00 carry an average of 12.41 additional riders per trip (95% CI: 8.17, 16.64; $p < 0.001$), and trips at 7:00 carry 10.31 additional riders (95% CI: 7.39, 13.24; $p < 0.001$).

Trips operating at 8:00 ($\beta = 0.75$; 95% CI: $-1.23, 2.72$; $p = 0.459$) and 9:00 ($\beta = -0.57$; 95% CI: $-2.31, 1.17$; $p = 0.520$) do not differ significantly from the 10:00 reference period. Similarly, trips between 11:00 and 13:00 show no statistically significant differences relative to 10:00.

In contrast, modest increases in utilization are observed in the early afternoon. Trips at 14:00 carry 4.48 additional riders per trip (95% CI: 2.05, 6.91; $p < 0.001$), at 15:00 carry 3.10 additional riders (95% CI: 1.01, 5.20; $p = 0.004$), and at 16:00 carry 2.85 additional riders (95% CI: 0.83, 4.87; $p = 0.006$).

Evening service is associated with significantly lower utilization. Trips at 18:00 carry 4.71 fewer riders per trip (95% CI: $-6.46, -2.96$; $p < 0.001$), while trips at 20:00 carry 7.61 fewer riders (95% CI: $-9.39, -5.83$; $p < 0.001$), and trips at 21:00 carry 8.02 fewer riders (95% CI: $-11.08, -4.97$; $p < 0.001$), relative to the 10:00 reference hour.

Boardings per trip also vary systematically across days of the week (Table 3). Relative to Monday, trips on Tuesday carry 3.89 fewer riders per trip on average (95% CI: $-5.65, -2.14$; $p < 0.001$). Larger declines are observed midweek, with Wednesday trips carrying 6.70 fewer riders (95% CI: $-8.42, -4.99$; $p < 0.001$) and Thursday trips carrying 7.01 fewer riders (95% CI: $-8.80, -5.22$; $p < 0.001$).

Utilization declines further on Friday ($\beta = -12.45$; 95% CI: $-13.93, -10.97$; $p < 0.001$). Weekend service also exhibits lower average boardings per trip relative to Monday, with Saturday trips carrying 12.79 fewer riders (95% CI: $-14.40, -11.17$; $p < 0.001$) and Sunday trips carrying 9.73 fewer riders (95% CI: $-14.13, -5.33$; $p < 0.001$).

Only two routes exhibit significantly higher average boardings per trip than Route 14. Route 41 carries 7.62 additional riders per trip on average (95% CI: $2.42, 12.83$; $p = 0.004$), and Route 20 carries 4.63 additional riders (95% CI: $1.02, 8.23$; $p = 0.012$).

In contrast, most routes operate with significantly fewer boardings per trip relative to Route 14. Among the largest negative differences, Route 32 carries 14.47 fewer riders per trip (95% CI: $-21.33, -7.61$; $p < 0.001$), Route 27 carries 13.13 fewer riders (95% CI: $-16.27, -9.99$; $p < 0.001$), and Route 25 carries 12.53 fewer riders (95% CI: $-15.69, -9.37$; $p < 0.001$). Similarly, Routes 26 ($\beta = -10.27$; 95% CI: $-13.71, -6.84$; $p < 0.001$), 52 ($\beta = -10.95$; 95% CI: $-15.24, -6.66$; $p < 0.001$), and 9C ($\beta = -9.36$; 95% CI: $-13.32, -5.40$; $p < 0.001$) show substantial negative differences relative to the reference route.

Several additional routes exhibit moderate but statistically significant reductions in average boardings per trip, including Routes 2, 3, 4, 5, 7, 8, 9, 15, 24, and 30 (all $p \leq 0.004$). Two routes—Route 17 ($p = 0.770$) and Route 40 ($p = 0.180$)—do not differ significantly from Route 14.

Overall, these findings indicate persistent cross-route heterogeneity in average trip-level ridership that remains even after accounting for systematic differences by hour of day and day of week.

RQ 3:

Descriptive Analysis

Median boardings per trip vary substantially across routes (Table 4), ranging from 3 riders per trip to 14 riders per trip. The highest median utilization is observed on Magalia – Chico, with a median of 14 boardings per trip (mean = 29.44, SD = 38.15; 231 trips). This is followed by Chico – Oroville, with a median of 11 boardings per trip (mean = 25.12, SD = 36.66; 687 trips), and Park/MLK/Forest, with a median of 9 boardings per trip (mean = 26.83, SD = 37.78; 75 trips).

In contrast, the lowest median utilization is observed on Oro Dam, South Oroville, and Student Shuttle – Warner/Oak, each with a median of 3 boardings per trip. These routes also exhibit lower mean boardings and smaller dispersion relative to higher-utilization routes.

The majority of routes cluster within a relatively narrow band of median boardings between 5 and 7 riders per trip. This concentration suggests that while a small number of routes experience substantially higher typical trip-level demand, most routes operate within a moderate utilization range.

Table 4. Distribution of boardings per route

<i>Route Name</i>	<i>Median</i>	<i>Mean</i>	<i>SD</i>	<i>N</i>
<i>Magalia - Chico</i>	14.0	29.44	38.15	231
<i>Chico - Oroville</i>	11.0	25.12	36.66	687
<i>Park/MLK/Forest</i>	9.0	26.83	37.78	75
<i>Bruce / Manzanita</i>	8.0	13.96	18.39	288
<i>Oroville - Biggs</i>	7.0	11.94	13.29	160
<i>Cedar Loop</i>	7.0	16.32	30.79	93
<i>Thermalito</i>	7.0	13.69	23.26	233

<i>Esplanade/SR99</i>	7.0	14.57	20.87	630
<i>Chico Airport Express</i>	6.5	11.65	13.52	78
<i>First / East</i>	6.0	12.91	16.18	873
<i>Mangrove</i>	6.0	11.91	15.67	716
<i>E. 8th Street</i>	6.0	12.84	17.84	729
<i>Paradise - Chico</i>	6.0	14.77	23.96	198
<i>Park/Forest/MLK</i>	6.0	17.89	35.43	518
<i>Nord / East</i>	6.0	13.38	19.69	784
<i>Park/Fair/Forest</i>	5.0	13.82	23.44	217
<i>Student Shuttle - Nord</i>	5.0	10.31	19.12	163
<i>Gridley - Chico</i>	4.0	11.38	17.62	24
<i>Esplanade / Lassen</i>	4.0	11.44	21.06	982
<i>Olive Highway</i>	4.0	9.76	16.42	201
<i>Oro Dam</i>	3.0	6.64	9.26	186
<i>South Oroville</i>	3.0	5.66	7.62	151
<i>Student Shuttle - Warner/Oak</i>	3.0	7.39	12.12	276

Across routes, the standard deviation of boardings per trip is large relative to the median, particularly for higher-demand routes. For example, Magalia – Chico (median = 14.0; SD = 38.15), Chico – Oroville (median = 11.0; SD = 36.66), and Park/MLK/Forest (median = 9.0; SD = 37.78) exhibit substantial variability in ridership across trips. In each of these cases, the standard deviation exceeds or is several times larger than the median, indicating that while a typical trip carries a moderate number of riders, there are frequent instances of trips with much higher passenger counts.

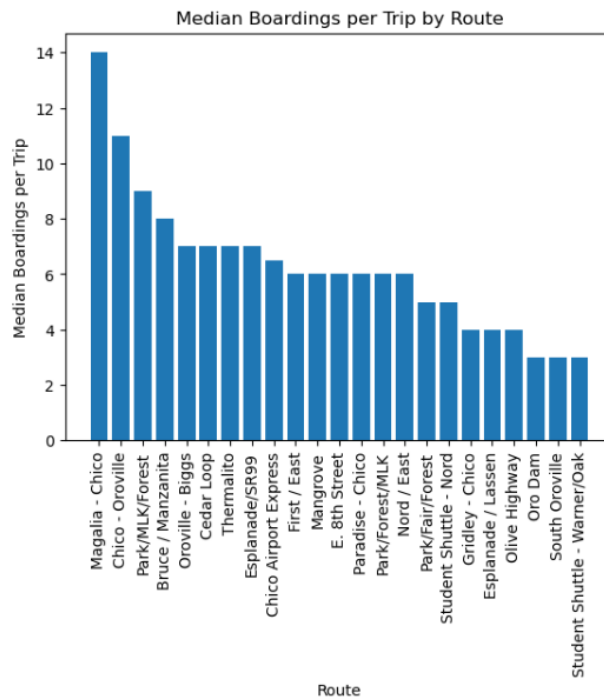
This pattern suggests considerable within-route variability, where ridership is not evenly distributed across trips but instead fluctuates based on time of day, day of week, or other operational factors. Even routes with moderate median values, such as Cedar Loop (median = 7.0; SD = 30.79) and Park/Forest/MLK (median = 6.0; SD = 35.43), show similarly high dispersion, reinforcing the presence of inconsistent demand within routes.

The bar chart of median boardings per trip (Figure 12) further illustrates the distribution of route-level utilization. Only a small number of routes—such as Magalia – Chico and Chico – Oroville—exceed 10 median riders per trip, indicating relatively higher and more consistent demand. In contrast, a large proportion of routes fall within the 4–7 median rider range, while several routes—including Oro Dam, South Oroville, and Student Shuttle – Warner/Oak—have median values of 3 or lower.

Overall, these results highlight a clear imbalance in ridership distribution across routes, with a few routes experiencing relatively higher typical demand and many others operating at lower median utilization levels. The combination of low median values and high variability suggests that demand is uneven not only across routes but also within routes, with certain trips contributing disproportionately to total ridership.

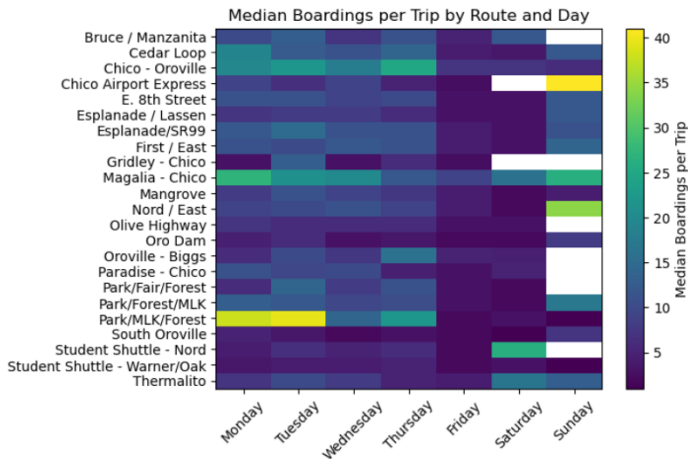
This pattern reflects heterogeneity in demand across the network and suggests that utilization is not uniformly distributed across routes.

Figure 12. Bar chart of median boardings per trip



The route-by-day heatmap (Figure 13) further reveals that many routes exhibit higher median boardings on weekdays compared to weekends, while some routes show more stable patterns across days. This variation suggests differences in route function and demand structure across the system.

Figure 13. Route by day heatmap



Modelling Results:

The Poisson model (Figure 14) was statistically significant, with all day-of-week indicators and most hourly indicators exhibiting strong associations with ridership ($p < 0.001$). Model fit statistics indicated that temporal variables explain a substantial portion of the variation in ridership (pseudo $R^2 = 0.953$). The Pearson chi-square statistic substantially exceeded the residual degrees of freedom, indicating overdispersion in the Poisson model and suggesting additional variability not captured by the included predictors. While this violates the Equi dispersion assumption, the model is retained for the purpose of generating expected ridership values used in route-level observed-to-expected comparisons. Because the primary objective of this analysis is relative comparison rather than coefficient-level inference, the impact of overdispersion on the study's conclusions is limited. However, results should be interpreted with appropriate caution.

Figure 14. Poisson Model

Generalized Linear Model Regression Results						
Dep. Variable:	boardings_per_trip	No. Observations:	8493			
Model:	GLM	Df Residuals:	8470			
Model Family:	Poisson	Df Model:	22			
Link Function:	Log	Scale:	1.0000			
Method:	IRLS	Log-Likelihood:	-93880.			
Date:	Tue, 17 Mar 2026	Deviance:	1.5602e+05			
Time:	16:03:58	Pearson chi2:	2.70e+05			
No. Iterations:	6	Pseudo R-squ. (CS):	0.9529			
Covariance Type:	nonrobust					
	coef	std err	z	P> z	[0.025	0.975]
Intercept	2.8591	0.013	221.090	0.000	2.834	2.884
C(dow, Treatment(reference='Monday'))[T.Friday]	-0.8793	0.009	-93.754	0.000	-0.898	-0.861
C(dow, Treatment(reference='Monday'))[T.Saturday]	-1.0439	0.014	-74.724	0.000	-1.071	-1.017
C(dow, Treatment(reference='Monday'))[T.Sunday]	-0.4628	0.028	-16.436	0.000	-0.518	-0.408
C(dow, Treatment(reference='Monday'))[T.Thursday]	-0.2169	0.010	-22.206	0.000	-0.236	-0.198
C(dow, Treatment(reference='Monday'))[T.Tuesday]	-0.0862	0.008	-10.476	0.000	-0.102	-0.070
C(dow, Treatment(reference='Monday'))[T.Wednesday]	-0.1772	0.009	-19.224	0.000	-0.195	-0.159
C(hour, Treatment(reference=10))[T.5]	1.0810	0.038	28.571	0.000	1.007	1.155
C(hour, Treatment(reference=10))[T.6]	0.7757	0.017	45.666	0.000	0.742	0.809
C(hour, Treatment(reference=10))[T.7]	0.5232	0.015	35.680	0.000	0.494	0.552
C(hour, Treatment(reference=10))[T.8]	0.0501	0.016	3.166	0.002	0.019	0.081
C(hour, Treatment(reference=10))[T.9]	-0.0276	0.016	-1.683	0.092	-0.060	0.005
C(hour, Treatment(reference=10))[T.11]	-0.0295	0.017	-1.752	0.080	-0.062	0.003
C(hour, Treatment(reference=10))[T.12]	0.0723	0.017	4.324	0.000	0.040	0.105
C(hour, Treatment(reference=10))[T.13]	0.1605	0.016	9.814	0.000	0.128	0.193
C(hour, Treatment(reference=10))[T.14]	0.2983	0.015	19.318	0.000	0.268	0.329
C(hour, Treatment(reference=10))[T.15]	0.1682	0.015	10.963	0.000	0.138	0.198
C(hour, Treatment(reference=10))[T.16]	0.2532	0.015	16.850	0.000	0.224	0.283
C(hour, Treatment(reference=10))[T.17]	0.1324	0.015	8.645	0.000	0.102	0.162
C(hour, Treatment(reference=10))[T.18]	-0.3558	0.019	-18.598	0.000	-0.393	-0.318
C(hour, Treatment(reference=10))[T.19]	-0.4003	0.023	-17.488	0.000	-0.445	-0.355
C(hour, Treatment(reference=10))[T.20]	-0.6258	0.031	-20.422	0.000	-0.686	-0.566
C(hour, Treatment(reference=10))[T.21]	-0.0445	0.042	-1.058	0.290	-0.127	0.038

The intercept corresponds to expected ridership for trips operating at 10:00 on Monday. Exponentiating the intercept ($\beta = 2.8591$) yields an expected value of approximately 17.4 riders per trip for this reference category.

Boardings per trip vary significantly across days of the week relative to Monday. Friday trips have approximately 58.5% lower ridership ($(\exp(-0.8793) - 1) \times 100$; $p < 0.001$). Similarly, Saturday shows a 64.8% decrease ($\beta = -1.0439$; $p < 0.001$), while Sunday shows a 37.0% decrease ($\beta = -0.4628$; $p < 0.001$). Tuesday, Wednesday, and Thursday also exhibit statistically significant declines relative to Monday, with ridership lower by 8.3%, 16.3%, and 19.5%, respectively (all $p < 0.001$).

Substantial variation is also observed across hours of operation relative to the 10:00 reference period. Early morning trips exhibit significantly higher ridership. Trips at 5:00 show a 194.7% increase in ridership ($\beta = 1.0810$; $p < 0.001$), while trips at 6:00 show a 117.2% increase ($\beta = 0.7757$; $p < 0.001$). Trips at 7:00 have approximately 68.8% higher ridership ($\beta = 0.5232$; $p < 0.001$). In contrast, 8:00 shows only a modest 5.2% increase ($p = 0.002$), while 9:00 and 11:00 are not statistically different from 10:00.

Ridership increases again during the early afternoon. Trips at 12:00, 13:00, 14:00, 15:00, 16:00, and 17:00 show increases of approximately 7.5%, 17.4%, 34.7%, 18.3%, 28.8%, and 14.2%, respectively (all $p < 0.001$), indicating a secondary peak in ridership during these hours.

Evening service is associated with lower ridership. Trips at 18:00, 19:00, and 20:00 show decreases of approximately 29.9%, 33.0%, and 46.5%, respectively (all $p < 0.001$), indicating reduced utilization during later hours. The coefficient for 21:00 is not statistically significant, suggesting similar ridership to the 10:00 baseline after accounting for variability.

To identify underutilized routes, predicted ridership values were generated for each trip and compared with observed boardings. These values were aggregated to the route level, and an observed-to-expected utilization ratio was computed for each route. This ratio provides a standardized measure of route performance, where values below 1 indicate that a route carries fewer riders than expected given its operating time. Routes in the bottom 15th percentile of this ratio distribution were classified as underutilized and are presented in Table 5.

Table 5. Route utilization based on observed-to-expected ridership ratios (bottom 15th percentile).

Route Name	Avg. Observed Riders per Trip	Avg. Expected Riders per Trip	Number of Trips	Utilization Ratio (O/E)
<i>South Oroville</i>	5.66	14.83	151	0.38
<i>Oro Dam</i>	6.64	15.75	186	0.42
<i>Student Shuttle-Warner/Oak</i>	7.39	15.27	276	0.48
<i>Gridley – Chico</i>	11.38	22.15	24	0.51

Substantial variation in utilization ratios was observed across routes. Several routes consistently exhibited markedly lower ridership than expected. Among the most underutilized routes, South Oroville had an average of 5.66 riders per trip compared to an expected 14.83, corresponding to a utilization ratio of 0.38. Similarly, Oro Dam carried an average of 6.64 riders per trip relative to an expected 15.75 (ratio = 0.42), and Student Shuttle – Warner/Oak carried 7.39 riders compared to an expected 15.27 (ratio = 0.48).

Gridley – Chico also demonstrated underutilization, with an average of 11.38 riders per trip compared to an expected 22.15 (ratio = 0.51), although this route had a relatively small number of observed trips and should be interpreted with caution.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary

This project examined the performance of the B-Line public transit system, with the goal of identifying patterns in ridership, evaluating service utilization, and evaluating route performance. Public transportation systems often face challenges related to fluctuating demand, inefficient resource allocation, and limited data infrastructure, particularly in mid-sized communities. This study addressed these challenges by applying a data-driven analytical framework to historical ridership data, while recognizing that the available data primarily capture validated boarding events and may not fully represent all riders, such as those using paper or student passes.

The conceptual framework of the study is grounded in the idea that transit system performance can be understood through three key dimensions: temporal demand patterns, trip-level service utilization, and route-level efficiency. Accordingly, the study was designed around three research questions that examine (1) how ridership varies over time, (2) how utilization differs across service conditions, and (3) which routes operate below expected performance levels. While this framework provides a structured approach, it relies on observable operational variables and does not incorporate external factors such as weather conditions, special events, or rider demographics, which may also influence ridership behavior.

The results demonstrate that ridership in the B-Line system is strongly influenced by time-of-day, day-of-week, and seasonal factors, with pronounced peaks during weekday commuting hours and significantly lower demand on weekends. At the trip level, utilization varies widely, with higher ridership during peak hours and substantial differences across routes. At the route level, the observed-to-expected framework revealed that a subset of routes consistently operates below expected ridership levels, indicating potential inefficiencies in service allocation. However, the estimation of expected ridership is based on system-wide temporal patterns, which assumes comparable demand conditions across routes operating at similar times and may not fully capture route-specific characteristics such as geographic coverage or rider purpose.

Overall, this project shows that historical transit data can be effectively used to evaluate system performance and generate actionable insights. By integrating statistical modeling with descriptive analysis, the study provides a comprehensive understanding of demand patterns, utilization, and route-level performance within the B-Line system, while acknowledging that the findings should be interpreted in the context of the data and modeling limitations described above.

Conclusion

This study was motivated by the need for data-driven evaluation of public transit systems in mid-sized communities, where limited resources and the absence of real-time data often constrain effective service planning. As highlighted in prior research, challenges such as fluctuating ridership demand, inefficient service allocation, and lack of localized analysis create a gap between theoretical optimization approaches and practical

implementation. This project addressed that need by applying a structured analytical framework to historical B-Line data, demonstrating how meaningful insights can be generated without relying on complex real-time systems.

The findings validate the importance of examining transit performance through multiple dimensions. By analyzing temporal demand patterns, the study confirmed that ridership is highly dependent on time-of-day, day-of-week, and seasonal factors, reinforcing the need for time-sensitive service planning. Through trip-level analysis, the study showed that service utilization varies significantly across operating conditions, highlighting imbalances in how service is distributed throughout the day. Finally, the route-level analysis provided a standardized method for identifying underperforming routes by comparing observed ridership to expected demand. Specifically, an observed-to-expected ridership ratio was calculated for each route by dividing the average observed boardings per trip by the corresponding model-predicted values. Routes with ratios substantially below 1 were considered underperforming, as they consistently carried fewer riders than expected given overall system patterns.

Based on this criterion, routes such as South Oroville, Oro Dam, Student Shuttle – Warner/Oak, and Gridley – Chico exhibited the lowest observed-to-expected ratios, indicating that these routes carry significantly fewer riders relative to expected demand and may represent areas of lower utilization.

Importantly, this study responds to the identified gap in the literature by demonstrating that smaller transit systems can leverage historical data to inform operational decisions. The use of regression and count-based modeling techniques allowed

for the identification of systematic patterns and disparities, bridging the gap between theoretical models and practical, implementable insights. These findings provide actionable guidance for B-Line transit planning and operations. For instance, identifying routes with consistently low observed-to-expected ridership ratios enables planners to reassess service allocation, such as reducing frequency, adjusting schedules, or redesigning routes to better align with demand. Conversely, routes and time periods with higher-than-expected ridership can be prioritized for increased service or resource allocation to prevent overcrowding.

Additionally, the temporal patterns observed in ridership—such as higher demand during midweek and specific months—can inform more efficient scheduling by concentrating service during peak demand periods and scaling back during consistently low-demand times, such as weekends or summer months. These insights also support data-driven decision-making for long-term planning, including route restructuring, targeted service improvements, and more effective use of operational resources. Overall, the analysis provides a framework for optimizing service delivery while maintaining accessibility and responsiveness to rider demand. Overall, the study confirms that a data-driven approach is both necessary and effective for improving transit system performance.

Recommendation

Based on the findings and overall project experience, several recommendations are proposed for future research and analytical practice.

From a content perspective, future studies can be enhanced by incorporating additional data sources such as academic calendars, different bus passes data and

demographic information to better understand factors influencing ridership. Including operational metrics such as bus capacity and on-time performance would also provide a more comprehensive evaluation of service effectiveness.

Future analyses of B-Line transit data should prioritize aligning research questions with appropriate analytical methods rather than relying on a single modeling approach. Different operational questions—such as demand forecasting, route-level evaluation, and temporal variation—require distinct analytical techniques to produce meaningful and reliable insights.

In addition, continued investment in data quality is essential. Ensuring consistent data collection, cleaning, and standardization across datasets will improve the accuracy and reliability of future analyses. Establishing standardized data pipelines or validation procedures would further support ongoing monitoring efforts.

Finally, B-Line should emphasize clear communication of analytical results through effective visualizations and structured reporting. Presenting findings in an accessible and interpretable format will enable planners and decision-makers to translate data-driven insights into actionable service improvements, such as adjusting schedules, reallocating resources, or refining route design.

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APPENDICES

Appendix 1:

CODE:

```
df['hour'] = df['timestamp'].dt.hour
df['day_of_week'] = df['timestamp'].dt.day_name()
df['month'] = df['timestamp'].dt.month

# Map months to seasons
season_map = {
    12: 'Winter', 1: 'Winter', 2: 'Winter',
    3: 'Spring', 4: 'Spring', 5: 'Spring',
    6: 'Summer', 7: 'Summer', 8: 'Summer',
    9: 'Fall', 10: 'Fall', 11: 'Fall'}
df['season'] = df['month'].map(season_map)
```

Appendix 2:

CODE:

```
# Number of unique trips in the dataset
validation["trip_id"].nunique()

# Inspect sample records to see trip_id, route, and time
validation[["trip_id", "route_id",
"timestamp"]].dropna().head(10)

# Number of boarding events per trip
validation.groupby("trip_id").size().describe()
```

Appendix 3:

CODE:

```
df['hour'] = df['timestamp'].dt.hour
df['day_of_week'] = df['timestamp'].dt.day_name()
```

Appendix 4:

CODE

```
trip_meta = (
    validation.sort_values("service_datetime")
    .groupby("trip_id", as_index=False)
    .agg(
        route_id=("route_id", "first"),
        hour=("hour", "first"),
        day_of_week=("day_of_week", "first"),
        trip_start_time=("service_datetime", "first")))
```